



Immunochemistry & Oncogenesis

MBBS BIOCHEMISTRY · PAPER 1

Immunoglobulins

IgG Structure

T & B Cell Immunity

Hypersensitivity

Complement System

Graft Rejection

Monoclonal Antibodies

Oncogenes

Tumour Suppressor Genes

p53 & Cell Cycle

Tumour Markers

Carcinogens

Apoptosis

Signal Transduction

Hormone Receptors

MASTER PYQ INDEX (2011–2025)

2011

2013

2015

2016

2017

2018

2019

2021

2023

2025

 EXAM PRIORITY MAP & USAGE STRATEGY

● **TIER 1 (High Yield — Must Master):** Immunoglobulin Structure & Classes (s1, s2), Hypersensitivity Reactions (s4), Cell Cycle & Regulators (s7), Oncogenes & Proto-oncogenes (s8), Tumour Suppressor Genes (s9), Tumour Markers (s11). Essential passing concepts.

● **TIER 2 (Medium Yield — Review):** T & B Cell Immunity (s3), Complement System (s5), Apoptosis (s10), Carcinogens & Cancer Biology (s12), Signal Transduction (s13), Hormone Receptor Classification (s14). Secondary pathway mechanisms.

● **TIER 3 (Low Yield — Collapsed on Screen):** Monoclonal Antibodies (s6), Essay Question Templates (s16). Fast read, high-level clinical methodologies.

00 Last Day Revision Mode

ULTRA-
CONCISE
EXAM
FACTS

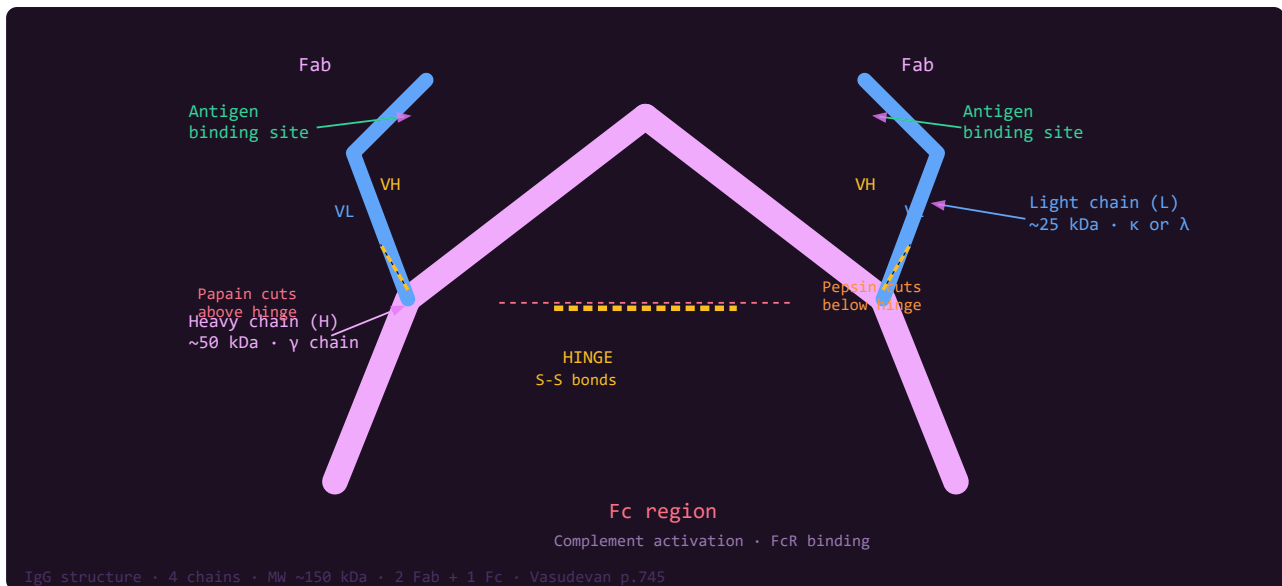
- ♦ **Immunoglobulins:** IgG (most abundant, crosses placenta), IgA (secretory, dimers in tears/milk), IgM (pentamer, primary response), IgE (allergic reactions, Type I hypersensitivity), IgD (B-cell surface marker).
- ♦ **Hypersensitivity:** Type I (Anaphylactic, IgE mediated), Type II (Cytotoxic, IgG/IgM + complement, e.g., Rh incompatibility), Type III (Immune-complex, e.g., SLE, glomerulonephritis), Type IV (Delayed-type, T-cell mediated, e.g., Mantoux test, contact dermatitis).
- ♦ **Cell Cycle Checkpoints:** G1/S (regulated by Cyclin D-CDK4/6 and Rb), G2/M (regulated by Cyclin B-CDK1). p53 acts at G1/S to trigger repair or apoptosis via Bax/Bcl2.
- ♦ **Oncogenes vs TSGs:** Proto-oncogenes need one mutation (gain of function, e.g., Ras, Myc, ErbB2) to promote cancer. Tumour Suppressors need two mutations (loss of function, 'Two-hit hypothesis', e.g., Rb, p53, BRCA1).
- ♦ **Tumour Markers:** AFP (Hepatocellular carcinoma, Yolk sac tumor), CEA (Colorectal cancer), PSA (Prostate cancer), CA-125 (Ovarian cancer), CA-19-9 (Pancreatic cancer).
- ♦ **Signal Transduction:** cAMP pathway (GPCR → Adenyl Cyclase → cAMP → PKA), IP3/DAG pathway (GPCR → Phospholipase C → IP3/DAG → Ca²⁺ release & PKC), RTK pathway (GF binding → dimerisation → autophosphorylation → Ras-Raf-MEK-ERK).

01 Immunoglobulin Structure

- ⚡ Immunoglobulins (Ig) = glycoproteins secreted by plasma cells (differentiated B lymphocytes). They are the humoral arm of adaptive immunity — specifically recognise and bind antigens.

BASIC IMMUNOGLOBULIN UNIT — IGG
PROTOTYPE

★ PYQ 2011-S, 2015-S, 2017-S, 2018, 2019, 2021, 2023



REGION	CHAINS	FUNCTION	KEY DETAIL
Fab (Fragment antigen binding)	VH + VL + CH1 + CL	Antigen binding (2 per IgG molecule)	Variable regions (VH, VL) contain HVR (hypervariable regions / CDRs) — actual antigen contact sites
Fc (Fragment crystallisable)	CH2 + CH3 (of both H chains)	Effector functions — complement, FcR binding, neonatal transport	Glycosylated at Asn-297 of CH2. Determines Ig class (IgG, IgA, IgM etc.)
Hinge region	Between CH1 and CH2	Flexibility for both Fab arms to bind antigen simultaneously	Contains inter-heavy-chain disulphide bonds. Absent in IgM, IgE
Variable region (V)	N-terminal of both H and L chains	Antigen specificity	3 CDRs (Complementarity Determining Regions) per V domain = paratope
Constant region (C)	Rest of H and L chains	Determines Ig class + effector functions	C region of H chain = isotype ($\gamma \rightarrow$ IgG; $\mu \rightarrow$ IgM; $\alpha \rightarrow$ IgA; $\delta \rightarrow$ IgD; $\epsilon \rightarrow$ IgE)

PAPAIN VS PEPSIN CLEAVAGE PYQ 2023

ENZYME	CLEAVAGE SITE	PRODUCTS	SIGNIFICANCE
Papain	Above hinge (N-terminal side)	2 × Fab (monovalent, cannot cross-link) + 1 × Fc	Used to study antigen binding separately from effector function
Pepsin	Below hinge (C-terminal side)	1 × F(ab') ₂ (bivalent, CAN cross-link antigen) + pFc' fragments (degraded)	F(ab') ₂ used therapeutically — retains binding + cross-linking ability without Fc-mediated complement activation

02 Ig Classes & Functions

FIVE IG CLASSES – PROPERTIES

★ PYQ 2017-S, 2019, 2025

CLASS	H CHAIN	STRUCTURE	SERUM %	KEY FUNCTIONS	SPECIAL FEATURE
IgG ★	γ (gamma)	Monomer	75–80%	Most abundant. Secondary response. Crosses placenta (only Ig to do so — neonatal immunity). Fixes complement. ADCC (NK cells). Opsonisation	4 subclasses (IgG1–4). IgG1 and IgG3 fix complement best
IgM ★	μ (mu)	Pentamer (J chain links 5 monomers)	5–10%	First Ig produced in primary response. Best complement activator (classical pathway). ABO blood group antibodies. Most efficient agglutinin (10 antigen binding sites)	Largest Ig (~900 kDa). Stays intravascular (large size). Raised in M-protein of Waldenström's macroglobulinaemia
IgA	α (alpha)	Monomer (serum) or Dimer (secretory, with J chain + secretory component)	10–15%	Mucosal immunity — secretory IgA (sIgA) in saliva, tears, colostrum, GI secretions, respiratory tract. First line mucosal defence	Secretory component added by epithelial cells. Protects from proteases. Colostrum IgA provides neonatal gut immunity
IgD	δ (delta)	Monomer	<1%	B cell surface receptor (antigen recognition on naïve B cells). Activates B cells	No known secreted effector function. Co-expressed with IgM on naïve B cells
IgE	ε (epsilon)	Monomer	Trace	Binds high-affinity FcεRI on mast cells and basophils → Type I hypersensitivity (allergy, anaphylaxis). Anti-parasitic (helminth) defence	Lowest serum concentration but highest affinity for FcR. Raised in atopy, parasitic infection. Target of omalizumab (anti-IgE therapy in asthma)

IG CLASS MNEMONIC

"GAMED" = IgG (most), IgA (mucosal), IgM (macro pentamer, 1st), IgE (eosinophil/allergy), IgD (differentiation)

"Only IgG crosses the placenta" — remember 'G' for 'Goes to baby'



MULTIPLE MYELOMA – IGG/IGA RELATED DISORDER

PYQ 2012-S

Malignant proliferation of a single plasma cell clone → secretes one type of Ig (M-protein/paraprotein) → monoclonal band on serum electrophoresis (M-spike in γ region). IgG most common (~55%), IgA next. Features: bone pain (osteolysis), anaemia, hypercalcaemia, renal failure (Bence-Jones protein = free light chains in urine), susceptibility to infection. Bence-Jones protein = free κ or λ light chains that pass through glomerular filter (small size).

03

T & B Cell Immunity

INNATE VS ADAPTIVE IMMUNITY

★ PYQ 2024

FEATURE	INNATE IMMUNITY	ADAPTIVE (ACQUIRED) IMMUNITY
Specificity	Non-specific — pattern recognition (PAMPs via TLRs)	Highly specific — antigen-specific receptors (BCR, TCR)
Memory	No immunological memory	Memory cells formed → faster, stronger secondary response
Response time	Immediate (minutes–hours)	Delayed (days–weeks for primary; faster secondary)
Components	Physical barriers (skin), phagocytes (neutrophils, macrophages), NK cells, complement, acute phase proteins, interferons	B lymphocytes (humoral) → antibodies. T lymphocytes (cellular) → cytotoxic, helper, regulatory
Diversity	Limited (~1000 PRR patterns)	Enormous (~ 10^{18} different antigen specificities)
MHC	Not MHC-restricted	MHC-restricted — TCR recognises antigen only when presented by MHC

B CELL VS T CELL MEDIATED IMMUNITY

★ PYQ 2021

B Cell (Humoral) Immunity:

- ✦ B cells mature in **Bone marrow**
- ✦ Activated by antigen (T cell help usually needed)
- ✦ Differentiate into **Plasma cells** (antibody-secreting) + Memory B cells
- ✦ Antibodies mediate: Neutralisation, Opsonisation, ADCC, Complement activation,

T Cell (Cellular) Immunity:

- ✦ T cells mature in **Thymus**
- ✦ Subtypes: **CD4⁺ Helper T** (Th1, Th2, Th17, Treg), **CD8⁺ Cytotoxic T**
- ✦ CD4 recognises antigen on **MHC II** (APCs); CD8 on **MHC I** (any nucleated cell)
- ✦ **Th1**: Activates macrophages, cytotoxic response, IFN- γ

Agglutination

♦ Surface marker: **CD19, CD20, CD21, BCR (surface Ig)**

♦ Best against: Extracellular pathogens, bacteria, toxins, viruses (before cell entry)

♦ **Th2:** B cell help, IgE, eosinophils, IL-4, IL-5, IL-13 (allergy/parasites)

♦ CTL (CD8⁺): Kills virus-infected cells via perforin/granzyme + Fas-FasL

♦ Best against: Intracellular pathogens, viruses, fungi, tumours

**GRAFT REJECTION TYPES****PYQ 2023**

TYPE	TIME	MECHANISM	PATHOLOGY
Hyperacute	Minutes to hours	Pre-existing antibodies (ABO incompatibility, anti-HLA)	Complement + coagulation → thrombosis → ischaemic necrosis
Acute	Days to weeks	T cell mediated (cellular) + antibody-mediated (humoral)	CD8 ⁺ CTL kill graft cells; CD4 ⁺ activate macrophages. Lymphocytic infiltrate
Chronic	Months to years	Both T cell and antibody; sustained immune response	Fibrosis, intimal thickening, narrowing of vessels → graft failure

04**Hypersensitivity Reactions****GELL & COOMBS CLASSIFICATION****★ PYQ 2022, 2025**

TYPE	NAME	MEDIATOR	TIME	MECHANISM	EXAMPLE
Type I ★	Immediate / Anaphylactic	IgE → mast cells/basophils	Minutes	Allergen → cross-links IgE on mast cells → degranulation → histamine, leukotrienes, prostaglandins → vasodilation, bronchoconstriction, ↑ vascular permeability	Anaphylaxis, asthma, urticaria, food allergy, atopic dermatitis
Type II	Cytotoxic / Antibody-mediated	IgG / IgM against cell surface	Hours	Antibody binds cell surface antigen → complement activation (MAC) + ADCC → cell lysis. Or blocking (myasthenia gravis)	Autoimmune haemolytic anaemia, Goodpasture's syndrome, Graves' disease, Myasthenia gravis, Haemolytic disease of newborn (Rh)
Type III	Immune Complex / Serum sickness	IgG/IgM-antigen complexes	Hours to days	Antigen-antibody complexes deposit in vessel walls/tissues → complement activation → inflammation, PMN recruitment → tissue damage	SLE, rheumatoid arthritis, post-streptococcal glomerulonephritis, serum sickness, Farmer's lung (extrinsic allergic alveolitis)
Type IV ★	Delayed-type / Cell-mediated	T lymphocytes (CD4 ⁺ Th1)	48–72 hours (delayed!)	Sensitised T cells release cytokines → activate macrophages → granuloma formation. No antibodies involved	TB (PPD test), contact dermatitis (poison ivy, Ni allergy), graft rejection, Coeliac disease, Type 1 DM

► TYPE I HYPERSENSITIVITY – MECHANISM IN DETAIL

PYQ 2022

Sensitisation Phase (first exposure) Allergen → APCs → Th2 cells → IL-4, IL-13 → B cells class-switch to IgE production → IgE binds FcεRI on mast cells/basophils (sensitisation, no symptoms)

↓ Second (or subsequent) allergen exposure

Elicitation Phase (re-exposure) Allergen cross-links IgE on mast cells → FcεRI clustering → **Degranulation** (within seconds-minutes):
 Preformed mediators: **Histamine** (vasodilation, pruritus, bronchoconstriction), **Tryptase** (mast cell marker)
 Newly synthesised: **Leukotrienes** C4, D4, E4 (potent bronchoconstrictors — "slow-reacting substances"), **PGD₂**, **PAF**
 Cytokines (late phase): IL-4, IL-5, TNF-α → eosinophil recruitment (late-phase reaction, 6–12 hours)

↓ Clinical manifestations

Systemic anaphylaxis: ↓ BP (vasodilation + ↑ permeability → hypovolaemia) + Bronchospasm + Urticaria. Treatment: **Epinephrine** (adrenaline) 1:1000 IM — reverses all components

05

Complement System

COMPLEMENT PATHWAYS – OVERVIEW

▶ THREE ACTIVATION PATHWAYS → COMMON TERMINAL PATHWAY

Classical Pathway **IgM or IgG** + antigen → C1q binds Fc → C1r+C1s activated → C4 + C2 → **C3 convertase (C4b2a)**

+

Lectin Pathway Mannose-Binding Lectin (MBL) binds mannose on pathogens → MASP → same C4/C2 route → **C3 convertase**

+

Alternative Pathway Spontaneous C3 hydrolysis (tickover) + Factors B, D, Properdin → **C3 convertase (C3bBb)**

↓ All converge → C3 cleavage (pivotal step)

C3 Cleavage C3 convertase cleaves C3 → **C3a** (anaphylatoxin — vasodilation, mast cell degranulation) + **C3b** (opsonin — binds CR1 on phagocytes)
C3b joins convertase → C5 convertase → **C5a** (potent anaphylatoxin + chemotaxin) + **C5b**

↓ C5b initiates

Membrane Attack Complex (MAC): C5b-6-7-8-9 Polymerised C9 inserts into cell membrane → pore → osmotic lysis
Deficiency of C5–C9 → susceptibility to *Neisseria* (meningococcal disease)

● COMPLEMENT FUNCTIONS MNEMONIC

"CLAMO" = Chemotaxis (C5a), Lysis (MAC), Anaphylatoxin (C3a, C5a), Mast cell degranulation, Opsonisation (C3b)

06

Monoclonal Antibodies

Expand ● TIER 3: Monoclonal Antibodies Details

HYBRIDOMA TECHNOLOGY – KÖHLER & MILSTEIN 1975

★ PYQ 2010-S, 2011, 2014

► MONOCLONAL ANTIBODY PRODUCTION

Step 1 — Immunise mouse Inject mouse with antigen of interest → B cell response → spleen contains antigen-specific B cells



Step 2 — Fusion (Cell fusion with PEG) Spleen B cells + Myeloma cells (HAT-sensitive, immortal) → **Hybridoma cells**
Selection in HAT medium (hypoxanthine, aminopterin, thymidine) — only hybridomas survive (myeloma dies, B cells die)

↓ Screen hybridoma supernatants for desired antibody

Step 3 — Clone & Expand Positive hybridoma cloned → each clone produces identical antibody → **Monoclonal antibody**

↓ Characterise antibody

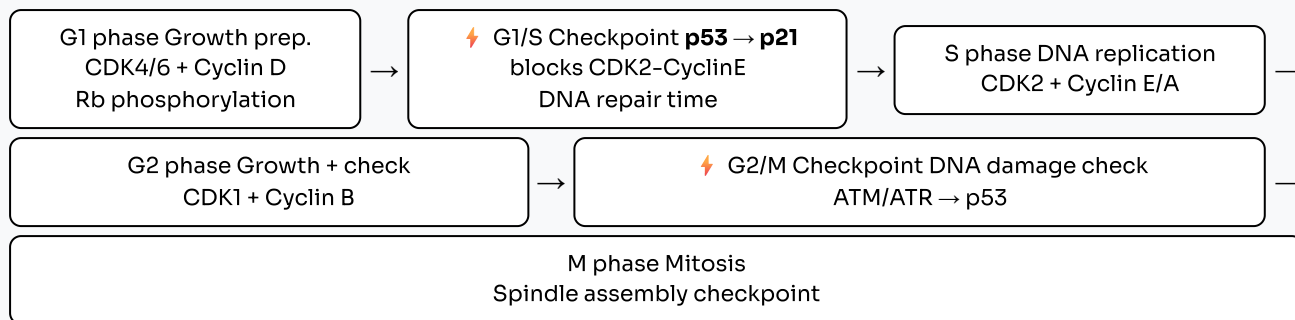
Step 4 — Humanise (for therapeutic use) Mouse mAbs → chimeric (mouse Fv + human Fc) → humanised (only CDRs murine) → fully human (phage display). Suffix: -omab (murine), -ximab (chimeric), -zumab (humanised), -umab (fully human)

THERAPEUTIC MONOCLONAL ANTIBODIES — KEY EXAMPLES

DRUG	TARGET	USE
Trastuzumab (Herceptin)	HER2 (ErbB2) receptor	HER2+ breast cancer — blocks HER2 signalling → ↓ proliferation
Rituximab	CD20 (B cells)	Non-Hodgkin lymphoma, rheumatoid arthritis — depletes B cells
Cetuximab	EGFR	Colorectal cancer, head & neck cancer
Bevacizumab (Avastin)	VEGF	Anti-angiogenic — colorectal, lung, renal cell carcinoma
Pembrolizumab	PD-1	Immune checkpoint inhibitor — melanoma, lung cancer
Omalizumab	IgE	Severe allergic asthma — prevents IgE binding to mast cells
Infliximab	TNF-α	Rheumatoid arthritis, Crohn's disease

07 Cell Cycle & Regulators

▶ CELL CYCLE PHASES & CHECKPOINTS



CELL CYCLE REGULATORS

★ PYQ 2010, 2021

REGULATOR	TYPE	FUNCTION	CANCER RELEVANCE
Cyclins (A, B, D, E)	Regulatory subunit — oscillate with cell cycle	Activate CDKs at specific cell cycle phases	Cyclin D overexpressed in many cancers (BCL1 gene amplification)
CDKs (CDK1, 2, 4, 6)	Catalytic kinase subunit — constant level	Phosphorylate Rb when bound to cyclin → cell cycle progression	CDK4/6 inhibitors (palbociclib) — breast cancer treatment
CKIs (CDK Inhibitors)	Inhibitory — p21(CIP1), p27(KIP1), p16(INK4a)	Block CDK activity → cell cycle arrest	p16 silenced by methylation in many cancers. p21 induced by p53
Rb (Retinoblastoma protein)	Tumour suppressor — "master brake"	Hypophosphorylated Rb sequesters E2F (transcription factor) → no S phase genes. CDK phosphorylates Rb → E2F released → cell enters S phase	Both copies lost → Retinoblastoma. Knudson's "two-hit hypothesis"
p53	Tumour suppressor — "guardian of genome"	DNA damage → p53 stabilised → p21 (arrest) or BAX/PUMA (apoptosis)	Mutated in ~50% of all human cancers

08 Oncogenes & Proto-oncogenes

⚡ Proto-oncogene = normal gene that regulates cell growth, proliferation, differentiation.
Oncogene = mutated/overexpressed proto-oncogene that drives uncontrolled proliferation.

Proto-oncogenes are "accelerators" — in cancer, the accelerator is stuck down.

MECHANISMS OF PROTO-ONCOGENE → ONCOGENE CONVERSION

★ PYQ 2014-S, 2018, 2021, 2024

MECHANISM	HOW IT WORKS	EXAMPLE
Point mutation	Single base change in coding sequence → constitutively active protein	RAS gene point mutation (G12V, G12D) → Ras GTPase stuck in GTP-bound active state → continuous proliferation signals. Found in 30% of all human cancers
Gene amplification	Multiple copies of proto-oncogene → overexpression of protein	N-MYC amplification → neuroblastoma. HER2/neu amplification → breast cancer. DHFR amplification → methotrexate resistance
Chromosomal translocation	Gene moved next to strong promoter OR fusion gene created	t(9;22) = Philadelphia chromosome → BCR-ABL fusion → constitutive tyrosine kinase → CML [PYQ 2023]. t(8;14) = c-MYC next to IgH promoter → Burkitt's lymphoma. t(11;22) = EWS-FLI1 fusion → Ewing's sarcoma [PYQ 2024]
Insertional mutagenesis	Retroviral LTR inserts near proto-oncogene → ↑ transcription	Mouse mammary tumour virus (MMTV) activates Wnt1
Promoter rearrangement	Proto-oncogene brought under control of a strong promoter	Ig gene promoter fused to MYC (alternative to translocation)

CATEGORIES OF PROTO-ONCOGENE PRODUCTS

PYQ 2016-S, 2022

CATEGORY	EXAMPLES	NORMAL FUNCTION	CANCER
Growth factors	c-SIS (PDGF-B chain)	Stimulate cell proliferation from outside	Autocrine loop — tumour makes its own growth factor
Growth factor receptors (with RTK)	ERBB1 (EGFR), HER2 (ERBB2), RET, KIT, FLT3	Transduce growth signals → intracellular kinase cascade	Constitutively active → constant proliferation signal. HER2+ breast cancer → trastuzumab target
Signal transducers (G-proteins)	RAS (HRAS, KRAS, NRAS)	GTPase — turns signal on (GTP) and off (GDP) by hydrolysing GTP	Point mutation → cannot hydrolyse GTP → stuck "ON" → KRAS mutated in 90% pancreatic, 50% colorectal cancers
Non-receptor tyrosine kinases	ABL, SRC	Intracellular signalling	BCR-ABL (CML) → imatinib target. SRC → cancer cell invasion
Nuclear transcription factors	MYC, FOS, JUN	Regulate gene expression → cell growth genes	MYC overexpression → drives S-phase genes. N-MYC in neuroblastoma. c-MYC in Burkitt's
Cyclins / CDKs	Cyclin D1, CDK4	Cell cycle progression (as above)	Amplification → bypass G1/S checkpoint
Apoptosis inhibitors	BCL2	Prevents apoptosis (anti-apoptotic)	t(14;18) → BCL2 fused to IgH promoter → BCL2 overexpressed → B cells resist apoptosis → follicular lymphoma

ONCOGENE MNEMONIC

"RANK-MYC" = Ras, ABL-BCR(t9;22), N-MYC(neuroblastoma), K-ras(pancreas), MYC(Burkitt), Cyclin D

09 Tumour Suppressor Genes

⚡ TSGs = "brakes" of cell cycle. Both copies must be lost for tumour development (

Knudson's Two-Hit Hypothesis

). First hit = mutation; Second hit = LOH (loss of heterozygosity), deletion, methylation.

MAJOR TUMOUR SUPPRESSOR GENES

★ PYQ 2019-S, 2023, 2024

GENE	CHROMOSOME	NORMAL FUNCTION	CANCER WHEN LOST
p53 (TP53) ★ MOST ASKED	17p13.1	Transcription factor activated by DNA damage → p21 (arrest) → BAX/PUMA (apoptosis) → GADD45 (DNA repair). Keeps normal p53 at low levels via MDM2-mediated ubiquitination	Mutated in ~50% ALL cancers. Li-Fraumeni syndrome (germline) → diverse early-onset cancers
Rb (Retinoblastoma)	13q14	Sequesters E2F when hypophosphorylated → G1 arrest. "Master brake of cell cycle"	Both alleles lost → Retinoblastoma (hereditary bilateral vs sporadic unilateral). Also osteosarcoma, small cell lung cancer
BRCA1 / BRCA2	17q21 / 13q12	DNA repair (homologous recombination of DSBs). Genome integrity	Hereditary breast and ovarian cancer. BRCA1 → breast + ovarian + cervical. BRCA2 → breast (M+F) + pancreatic + prostate
APC	5q21	Regulates β -catenin (Wnt pathway). Promotes β -catenin degradation → prevents pro-proliferative gene expression	Both alleles lost → Familial Adenomatous Polyposis (FAP) → colorectal cancer. First hit in sporadic colorectal cancer too
PTEN	10q23	Phosphatase — dephosphorylates PIP_3 → opposes $PI3K$ → ↓ AKT → ↑ apoptosis	Cowden syndrome. Prostate, breast, endometrial cancers
VHL	3p25	Targets HIF-1 α for degradation. Controls angiogenesis via VEGF regulation	Von Hippel-Lindau syndrome → Renal cell carcinoma (clear cell), haemangioblastoma. Loss → HIF-1 α accumulates → VEGF → tumour vasculature

► P53 AS GUARDIAN OF THE GENOME

★ PYQ 2019-S, 2023, 2025

Normal state p53 kept low by **MDM2** (E3 ubiquitin ligase) → ubiquitinates p53 → proteasomal degradation. MDM2 is itself induced by p53 (negative feedback)

↓ DNA damage, oncogenic stress, hypoxia, ribosomal stress

Activation ATM/ATR kinases phosphorylate p53 at Ser15, Ser20 → MDM2 cannot bind → p53 stabilised + acetylated + tetramerises → TRANSCRIPTION FACTOR active

↓ p53 transcribes target genes — decision based on damage severity

Reversible damage → ARREST p21 (CDKN1A) → inhibits CDK2 → Rb stays unphosphorylated → E2F sequestered → no S phase → G1/S arrest → time for DNA repair

OR

Irreparable damage → APOPTOSIS BAX, PUMA, NOXA → mitochondrial pathway → cytochrome c release → caspase 9 → effector caspases → cell death

OR

Chronic/replicative stress → SENESCENCE Permanent cell cycle arrest. Prevents tumourigenesis in pre-malignant lesions

10

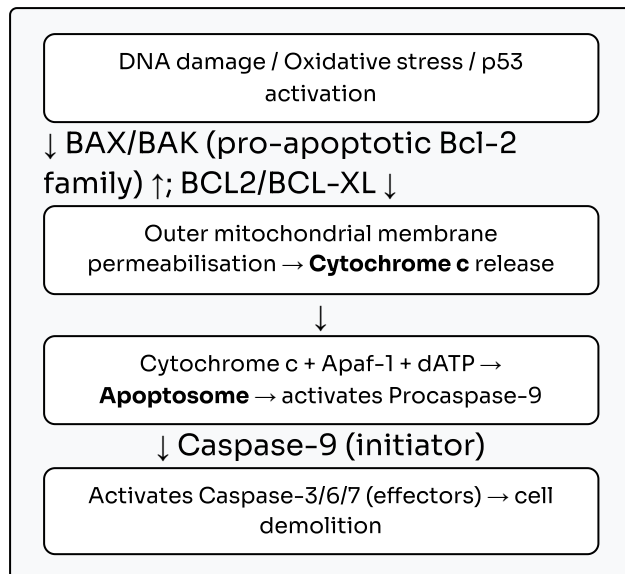
Apoptosis

- ⚡ Apoptosis = programmed cell death. Characterised by: cell shrinkage, chromatin condensation, membrane blebbing, DNA laddering (180 bp fragments), apoptotic bodies (phagocytosed — NO inflammation). Distinct from necrosis (swelling, lysis, inflammation).

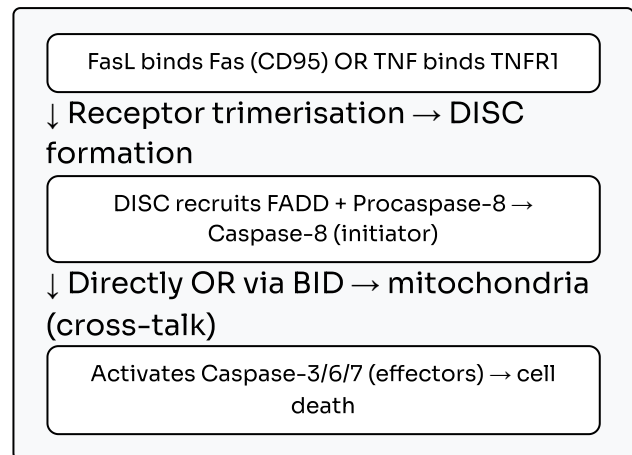
TWO PATHWAYS OF APOPTOSIS

★ PYQ 2023, 2024

Intrinsic (Mitochondrial) Pathway:



Extrinsic (Death Receptor) Pathway:



APOPTOSIS VS CANCER

PYQ 2023

BCL2 overexpression (t(14;18) in follicular lymphoma) → cells cannot undergo apoptosis → immortalised. **Survivin** (IAP family) overexpressed in many cancers → inhibits caspases. **Loss of p53** → cannot trigger apoptosis. **Cytochrome c also participates in ETC** — dual role (electron carrier + apoptosis trigger).

11

Tumour Markers

CLINICALLY IMPORTANT TUMOUR MARKERS

★ PYQ 2012-S, 2014, 2016-S, 2021, 2025

MARKER	TYPE	NORMAL	ASSOCIATED CANCER	SPECIAL NOTES
AFP (Alpha-fetoprotein)	Oncofetal (produced in fetal liver → re-expressed in tumour)	<20 ng/mL	Hepatocellular carcinoma (HCC), Germ cell tumours (yolk sac)	Also ↑ in pregnancy (normal), neural tube defects (maternal serum). Best marker for HCC monitoring
CEA (Carcinoembryonic antigen)	Oncofetal glycoprotein	<5 ng/mL (<2.5 in non-smokers)	Colorectal cancer (most useful), GI cancers, lung, breast	Not diagnostic (↑ in smokers, hepatitis). Used for monitoring recurrence after surgery. First oncofetal antigen described
PSA (Prostate Specific Antigen)	Serine protease (kallikrein family)	<4 ng/mL	Prostate cancer — most used marker	Produced only by prostate epithelium (organ-specific but NOT cancer-specific). ↑ in BPH, prostatitis too. Free PSA/total PSA ratio helps discriminate
CA-125	Mucin (MUC16)	<35 U/mL	Ovarian cancer (epithelial) — most useful	↑ in endometriosis, PID, peritonitis (non-specific). Used for monitoring response to chemotherapy
CA 19-9	Sialylated Lewis antigen	<37 U/mL	Pancreatic cancer (most sensitive), biliary tract cancer, gastric	Cannot produce it if Lewis antigen negative (5–10% of population) → false negative. Used for monitoring, not screening
CA 15-3	Mucin (MUC1)	<30 U/mL	Breast cancer (monitoring)	Not sensitive enough for early diagnosis. Used for monitoring metastatic breast cancer
β-hCG	Glycoprotein hormone	<5 mIU/mL (non-pregnant)	Choriocarcinoma, Gestational trophoblastic disease, Germ cell tumours (testicular)	Best marker for choriocarcinoma (one of most chemosensitive cancers). Also used for pregnancy testing
LDH	Enzyme (isoenzyme 1 in RBC, heart; 5 in liver)	140–280 U/L	Non-specific — lymphoma, testicular cancer, melanoma	Poor specificity. Correlates with tumour burden. LDH-1 elevated in myocardial infarction (LDH flip)
Calcitonin	Hormone (parafollicular C-cells)	<10 pg/mL	Medullary thyroid carcinoma (MTC)	Excellent marker for MTC. Used for diagnosis + monitoring. Screening of family members in MEN2
Thyroglobulin	Storage protein for thyroid hormones	<55 ng/mL	Differentiated thyroid carcinoma (papillary, follicular) — post-thyroidectomy	Should be undetectable after total thyroidectomy. Rise = recurrence

TUMOUR MARKERS MNEMONIC

"AFP Goes, CEA Comes, PSA Prostate, CA-125 Ovary, CA 19-9 Pancreas"

Oncofetal markers = AFP + CEA (both produced in fetal life, re-expressed in tumours)

12 Carcinogens & Cancer Biology

TYPES OF CARCINOGENS

★ PYQ 2023, 2024

TYPE	EXAMPLES	MECHANISM	CANCER
Chemical — Direct acting	Mustard gas, β -propiolactone, vinyl chloride	Directly alkylate DNA without metabolic activation	Bladder (vinyl chloride → angiosarcoma liver)
Chemical — Indirect (Procarcinogen)	Benzo[a]pyrene (tobacco smoke), Aflatoxin B1 (Aspergillus, mouldy groundnuts), Dimethylnitrosamine, β -Naphthylamine	Require metabolic activation by CYP450 → ultimate carcinogen → DNA adducts → mutations	Benzo[a]pyrene → lung (squamous). Aflatoxin B1 → HCC (especially with HBV). β -Naphthylamine → bladder
Physical — Radiation	UV-B (260–280 nm) → pyrimidine dimers; Ionising radiation (X-ray, γ -ray) → DSBs; α, β particles	DNA damage → mutations in TSGs and oncogenes	UV-B → skin cancer (XP patients). IR → leukaemia, thyroid cancer
Biological — Viruses (Oncogenic)	HBV, HCV (via HBsAg, insertional mutagenesis) · HPV 16, 18 (E6 → p53 degradation; E7 → Rb inactivation) · EBV (MYC translocation, t(8;14)) · HTLV-1 · HIV (indirectly)	Viral proteins inhibit TSGs or activate oncogenes; insertional mutagenesis; immunosuppression	HBV/HCV → HCC; HPV → cervical cancer, head & neck; EBV → Burkitt's lymphoma, NPC; HTLV-1 → ATL
Biological — Bacteria	H. pylori (CagA protein)	Chronic inflammation → MALT lymphoma, gastric adenocarcinoma	Gastric cancer, MALT lymphoma
Hormones	Oestrogen (prolonged excess), DES (diethylstilboestrol)	Mitogenic effect → ↑ cell proliferation → ↑ mutation acquisition chance	Endometrial cancer (oestrogen excess), clear cell carcinoma of vagina (DES exposure in utero)
Genetic	Hereditary mutations: BRCA1/2, APC, MLH1/MSH2 (Lynch), RB1, TP53 (Li-Fraumeni)	Germline first hit → somatic second hit → cancer at earlier age	Multiple organ cancers depending on gene

HALLMARKS OF CANCER (HANAHAH & WEINBERG)

1. Self-sufficiency in growth signals (oncogenes) · 2. Insensitivity to anti-growth signals (TSG loss) · 3. Evasion of apoptosis (BCL2↑, p53 lost) · 4. Unlimited replicative potential (telomerase↑) · 5. Sustained angiogenesis (VEGF↑, VHL loss) · 6. Tissue invasion and metastasis (MMP↑, E-cadherin↓) · 7. Reprogramming of energy metabolism (Warburg effect) · 8. Evading immune destruction (PD-L1↑)

Warburg effect: Cancer cells preferentially use aerobic glycolysis (glucose → lactate even in O₂) rather than OXPHOS → produces lactate + provides biosynthetic intermediates for rapid growth. Basis for FDG-PET scan (↑ glucose uptake in tumours).

13 Signal Transduction

G-PROTEIN COUPLED RECEPTOR (GPCR) PATHWAY

★ PYQ 2010, 2013, 2022, 2024

► CAMP SECOND MESSENGER SYSTEM – GLUCAGON/ADRENALINE

Step 1 — Ligand binding Hormone (glucagon/adrenaline) binds 7-TM GPCR → conformational change → GDP on Gα replaced by GTP → Gα-GTP dissociates from Gβγ

↓ Gα-GTP activates effector

Step 2 — Second messenger production Gas + Adenylyl Cyclase → ATP → **cAMP** (second messenger) + PPi
Gai inhibits adenylyl cyclase → ↓ cAMP (e.g., somatostatin, muscarinic M2)

↓ cAMP activates PKA

Step 3 — Protein Kinase A (PKA) cAMP binds regulatory subunits of PKA → releases catalytic subunits → active PKA phosphorylates target proteins (Ser/Thr):
Glycogen phosphorylase kinase (↑) → glycogenolysis ↑
Glycogen synthase (↓) → glycogenesis ↓
Lipase (HSL) (↑) → lipolysis ↑
CREB (nuclear) → gene transcription

↓ Signal terminated by

Gα GTPase activity: GTP → GDP → Gα reassociates with Gβγ → GPCR inactive
Phosphodiesterase (PDE): cAMP → 5'-AMP (theophylline, caffeine = PDE inhibitors → ↑ cAMP)



CHOLERA TOXIN – G-PROTEIN RELATED DISEASE

PYQ 2024

Cholera toxin (CT) ADP-ribosylates Gas at Arg201 → prevents GTP hydrolysis → Gas permanently active → adenyl cyclase permanently on → cAMP ↑↑↑ → PKA → CFTR opens → Cl⁻, HCO₃⁻ secretion → massive water secretion → "rice water" diarrhoea. Pertussis toxin ADP-ribosylates Gai → prevents Gi activation → no inhibition of cAMP → ↑ cAMP in airway → whooping cough.

PHOSPHATIDYLINOSITOL (PIP₂) – IP₃/DAG PATHWAY

PYQ 2013

Gαq activates **PLCβ** (or RTK activates PLCγ)

↓ PLCβ/γ cleaves PIP₂ (phosphatidylinositol-4,5-bisphosphate)

IP₃ (Inositol trisphosphate) Water-soluble → ER receptor → Ca²⁺ release from ER → ↑ cytosolic Ca²⁺

Ca²⁺ + Calmodulin → CaM-kinase → enzyme activation (myosin light chain kinase, phosphorylase kinase)

+

DAG (Diacylglycerol) Lipid-soluble → stays in membrane → activates **Protein Kinase C (PKC)** (also needs Ca²⁺)

PKC → Ser/Thr phosphorylation of target proteins

RECEPTOR TYROSINE KINASE (RTK) PATHWAY –
INSULIN/GROWTH FACTORS★ PYQ 2010-S, 2015-S,
2019-S, 2022

Step 1 Insulin (or EGF, PDGF, IGF-1) binds extracellular domain of RTK → receptor dimerises → **autophosphorylation** of Tyr residues on cytoplasmic domain

↓ Phospho-Tyr = docking sites for SH2-domain proteins

Step 2 — IRS-1 / Grb2-SOS / PI3K Insulin receptor: IRS-1 docked → PI3K recruited → PIP₂ → **PIP₃** → PDK1 → **AKT (PKB)** → glucose uptake (GLUT4), glycogen synthesis, protein synthesis, anti-apoptosis

Grb2-SOS → **RAS activated** → RAF → MEK → ERK → cell proliferation

↓ Signal terminated by

Protein tyrosine phosphatases (PTP1B) → remove Tyr phosphorylation. PTEN → removes PIP₃ → opposes PI3K



Insulin receptor = RTK (tyrosine kinase receptor) — dimer of 2α + 2β subunits linked by S-S bonds. Insulin binds α subunit → activates β-subunit Tyr kinase. This is distinct from steroid hormone receptors (nuclear) and GPCR (cAMP).

PYQ
2010-S

14

Hormone Receptor Classification

HORMONE CLASSIFICATION BY MECHANISM

★ PYQ 2017-S, 2024

TYPE	RECEPTOR LOCATION	SECOND MESSENGER	EXAMPLES	MECHANISM
Group I — Lipid soluble (Steroid, thyroid)	Intracellular (nuclear or cytoplasmic)	None — direct gene regulation	Cortisol, aldosterone, oestrogen, testosterone, progesterone, T3/T4, calcitriol	Hormone enters cell → binds nuclear receptor → receptor-hormone complex binds HRE (hormone response element) on DNA → activates/represses gene transcription
Group II — Peptide hormones / Catecholamines (GPCR)	Cell surface	cAMP, IP ₃ /DAG, Ca ²⁺	Glucagon, Adrenaline (β), FSH, LH, TSH, ACTH, ADH (V2), PTH, Calcitonin	Ligand → GPCR → Gs/Gi/Gq → adenylyl cyclase (cAMP) or PLC (IP ₃ +DAG) → PKA or PKC → phosphorylation cascade
Group IIb — RTK (Tyrosine kinase)	Cell surface (spans membrane)	PIP ₃ , RAS-ERK	Insulin, IGF-1, EGF, PDGF, FGF, NGF	Ligand → dimerisation → Tyr autophosphorylation → IRS-1/Grb2-SOS → PI3K-AKT and RAS-ERK cascades
Group IIc — JAK-STAT	Cytokine receptors (no intrinsic kinase)	STAT proteins	Growth hormone, Prolactin, Cytokines (interferons, IL-6), Erythropoietin, TPO	Ligand → receptor dimerises → associated JAK kinases phosphorylate STAT → STAT dimerises → enters nucleus → transcription
Group III — Guanylyl cyclase receptors	Cell surface (transmembrane guanylyl cyclase)	cGMP	ANP (atrial natriuretic peptide), BNP, NO (soluble GC)	ANP → membrane GC → GTP → cGMP → PKG → vasodilation, diuresis. NO → soluble GC → cGMP → smooth muscle relaxation (PDE5 inhibitors — sildenafil)

● LIPID-SOLUBLE HORMONE MNEMONIC

"STAVE" = Steroid hormones, Thyroid hormone, Aldosterone (mineralocorticoid), Vitamin D (calcitriol), Eicosanoids (some)

All lipid-soluble → cross membrane → nuclear receptor → direct gene transcription → effect in hours to days (vs peptide hormones: seconds to minutes)

15 Viva Q&A

IMMUNOGLOBULINS & IMMUNITY

Draw the structure of IgG and label all key regions. What is the difference between Fab and Fc?PYQ 2017-S, 2019, 2023

IgG structure: Y-shaped glycoprotein. MW ~150 kDa. 4 chains: 2 identical heavy (γ , ~50 kDa each) + 2 identical light (κ or λ , ~25 kDa each). Held together by disulphide bonds (inter-heavy in hinge region + heavy-light).

Fab (Fragment Antigen Binding):

- Contains: $VH + VL + CH1 + CL$ domains
- Function: Antigen binding (2 Fab arms per IgG = bivalent)
- Paratope = 3 CDRs (hypervariable loops) on each V domain
- Monovalent after papain cleavage → cannot agglutinate antigen

Fc (Fragment Crystallisable):

- Contains: $CH2 + CH3$ domains of both heavy chains
- Function: Effector functions — classical complement activation ($C1q$ binds $CH2$), $Fc\gamma R$ binding (NK cells, macrophages for ADCC), $FcRn$ binding (neonatal Fc receptor → transcytosis across placenta + prolonged half-life)
- Glycosylated at Asn-297

Papain: cuts above hinge → 2 Fab + 1 Fc. Pepsin: cuts below hinge → 1 $F(ab')_2$ (still bivalent) + degraded pFc'

Only IgG crosses the placenta — explain the mechanism.PYQ implied in classification Qs

The neonatal Fc receptor ($FcRn$) on placental syncytiotrophoblasts mediates **transcytosis** of IgG across the placental barrier.

- $FcRn$ binds IgG Fc region at acidic pH (in endosomes) after receptor-mediated endocytosis
- Transported across cell → released at neutral pH on fetal side → IgG enters fetal circulation
- This is the **ONLY** Ig small enough and with the right Fc for $FcRn$ binding
- IgM, IgA, IgE, IgD cannot cross (IgM too large = pentamer; others lack affinity for $FcRn$)
- Clinical significance: Maternal IgG protects neonate for first 3–6 months (maternal IgG half-life ~23 days)
- Danger: Maternal anti-Rh IgG crosses → haemolytic disease of the newborn (HDN)
- Also: maternal thyroid-stimulating Ig (IgG) crosses → neonatal hyperthyroidism

$FcRn$ also responsible for the long half-life of IgG (~23 days) — rescues IgG from lysosomal degradation in endothelial cells (same mechanism)

ONCOGENES & CANCER

What is p53? Why is it called the "Guardian of the Genome"?PYQ 2019-S, 2023, 2025

p53: Tumour suppressor protein (53 kDa) encoded by TP53 gene on chromosome 17p13. Normally kept at low levels by MDM2-mediated ubiquitination → proteasomal degradation.

Why "Guardian of the Genome":

- **Sensor:** Activated by DNA damage (ATM/ATR), oncogenic stress (ARF pathway), hypoxia, ribonucleotide depletion → phosphorylated → resistant to MDM2
- **G1/S checkpoint:** p53 → p21 (CIP1) → inhibits CDK2-Cyclin E → Rb stays unphosphorylated → E2F sequestered → S-phase genes not expressed → arrest
- **Apoptosis:** Irreparable damage → p53 → BAX, PUMA, NOXA (pro-apoptotic Bcl-2 family) → mitochondrial pathway → caspase activation → cell death
- **DNA repair:** p53 → GADD45, p48 (XPC partner) → nucleotide excision repair
- **Senescence:** Chronic activation → p21, p16 → permanent growth arrest

Cancer relevance: Mutated in ~50% all human cancers (most commonly mutated TSG). Germline p53 mutation → Li-Fraumeni syndrome (sarcomas, breast cancer, brain tumours at young age). Loss of p53 → cells cannot arrest or die → genomic instability → cancer progression.

"Without p53, damaged cells survive and accumulate further mutations → cancer. p53 is the last line of defence — hence Guardian of the Genome."

Explain Knudson's Two-Hit Hypothesis with retinoblastoma as an example.PYQ implied in TSG questions

Two-Hit Hypothesis (Alfred Knudson, 1971):

- Tumour suppressor genes require BOTH alleles to be inactivated for tumour development (recessive at cellular level)
- **Hereditary retinoblastoma:** First hit = germline mutation in one RB1 allele (inherited). Second hit = somatic mutation/deletion of remaining normal allele in retinal cell. Because first hit is in every cell → only needs one more somatic hit → bilateral, multiple tumours, early age
- **Sporadic retinoblastoma:** Both hits must occur somatically in the SAME retinal cell → much less likely → unilateral, single tumour, later age of onset
- **Why "Two-Hit":** Rb protein (pRb) is the cell cycle brake. One functional Rb is sufficient (haploinsufficiency not seen in most cases). Both copies must be lost to release the brake

This model applies to: BRCA1/2 (one germline + one somatic → cancer); APC (FAP); VHL (renal cell carcinoma); p53 in Li-Fraumeni

Cancer may be caused by excessive activity of protein tyrosine kinase — explain.PYQ 2022

Normal RTK signalling: Growth factor binds RTK → transient Tyr kinase activation → proliferation signal → Tyr phosphatase terminates signal.

Excessive Tyr kinase in cancer:

- **Mutation → constitutive activation:** RET point mutation (MEN2) → RET RTK always "on" → medullary thyroid carcinoma. KIT mutation → GIST (gastrointestinal stromal tumour) → imatinib target
- **Gene amplification:** HER2/neu amplification → excessive ErbB2 RTK → sustained RAS-ERK + PI3K-AKT → breast cancer cell proliferation, survival, metastasis
- **Fusion gene:** BCR-ABL (t(9;22)) → constitutively active non-receptor Tyr kinase → RAS-ERK + PI3K-AKT → CML. Imatinib (Gleevec) specifically inhibits BCR-ABL kinase → targeted therapy
- **Autocrine loop:** Tumour makes its own growth factor (PDGF) → activates its own RTK → self-sustaining proliferation

Key concept: Normal RTK = light switch (on/off). Cancer RTK = stuck "on" switch. Targeted therapies (imatinib, trastuzumab, gefitinib) are designed to flip it back off.

 SIGNAL TRANSDUCTION

Methotrexate is used in anti-cancer therapy — explain its mechanism. PYQ 2011, 2012, 2016-S, 2019-S

Target: Dihydrofolate Reductase (DHFR)

Mechanism:

- Folic acid → DHF → THF (by DHFR, requires NADPH)
- THF → one-carbon donors → thymidylate synthesis (dTTP), purine ring synthesis
- MTX = structural analogue of folic acid → binds DHFR with ~1000× higher affinity than natural substrate → **competitive inhibition** (but effectively irreversible at therapeutic concentrations)
- DHF cannot be reduced → THF pool depleted → no thymidylate (TTP) → no DNA synthesis → S-phase arrest → apoptosis

Selectivity: Rapidly dividing cells (cancer, immune cells) rely more on new nucleotide synthesis → more sensitive.

Rescue: Leucovorin (folinic acid = 5-formyl-THF) given after high-dose MTX → bypasses DHFR → rescues normal cells (cancer cells cannot take up leucovorin as well)

Uses: Leukaemia (ALL, CML), rheumatoid arthritis, psoriasis, Crohn's disease, ectopic pregnancy

Resistance mechanisms: ↑ DHFR expression (gene amplification), mutant DHFR with ↓ affinity, ↓ folate carrier, ↓ polyglutamation

Telomerase is involved in ageing and cancer — explain.PYQ 2016-S, 2025

Telomeres: TTAGGG repeats (2–15 kb) at chromosome ends. Protect chromosomes from end-degradation, fusion, and genomic instability.

End-replication problem: Normal DNA polymerase cannot replicate the very 3' end of lagging strand → telomeres shorten with each cell division (~50–200 bp/division).

Telomerase: Ribonucleoprotein enzyme (TERT = catalytic subunit, TERC = RNA template with AAUCCC sequence). Extends telomeres using its own RNA as template (reverse transcriptase activity).

Ageing:

- Somatic cells (no telomerase) → telomeres shorten → reach critical length → Hayflick limit → replicative senescence → contributes to ageing
- Stem cells + germ cells express low telomerase → partial protection
- Dyskeratosis congenita = telomerase mutations → premature ageing syndrome, aplastic anaemia

Cancer:

- Cancer cells reactivate TERT → telomeres maintained → unlimited replication potential → immortalisation
- ~85–90% of human cancers have telomerase activity
- Alternative Lengthening of Telomeres (ALT) = recombination-based → 10–15% of cancers
- TERT = potential anticancer drug target

Summary: Telomere shortening → ageing/senescence. Telomerase reactivation → cancer immortality. Two sides of the same coin.

16

Essay Question Templates

Expand ● TIER 3: Essay Question Templates Details

EQ – CANCER IS A GENE-RELATED DISEASE [PYQ 2011-S, 2024]

"Cancer is a gene-related disease."

- Definition:** Cancer = clonal expansion of a cell that has acquired multiple somatic mutations in genes controlling proliferation, differentiation, and apoptosis. It is fundamentally a disease of the genome.
- Two classes of cancer genes:** (a) Proto-oncogenes → Oncogenes: gain-of-function mutations → dominant, need only ONE allele mutated (accelerators stuck on). (b) Tumour Suppressor Genes: loss-of-function → recessive, need BOTH alleles lost (brakes cut). Both classes must be compromised for full malignancy.
- Genetic alterations:** Point mutations (RAS in pancreatic cancer), gene amplification (HER2 in breast cancer), chromosomal translocations (BCR-ABL → CML, MYC → Burkitt's lymphoma), deletions (RB1 → retinoblastoma), methylation silencing (APC in colorectal cancer), microsatellite instability (Lynch syndrome).
- Multi-step carcinogenesis:** Vogelstein's model (colorectal cancer): APC mutation → RAS activation → SMAD4 loss → p53 loss → invasive carcinoma. Each step = additional mutation. Typically requires 5–7 driver mutations accumulated over years.
- Inherited cancers:** Germline mutations (BRCA1/2, APC, RB1, TP53, MLH1/MSH2) → provide "first hit" in every cell → cancer at earlier age with defined tissue specificity.
- Epigenetics:** DNA methylation (silences TSGs), histone modification → heritable gene expression changes without DNA sequence change → also contribute to cancer development.
- Examiner tip:** Cancer requires MULTIPLE genetic changes (multi-step). No single gene mutation is sufficient. Hereditary cancers have germline first hit (fewer steps needed → earlier onset).

 EQ – PROTO-ONCOGENES ARE REGULATORY GENES [PYQ 2021]

"Proto-oncogenes are regulatory genes."

1. **Definition:** Proto-oncogenes = normal cellular genes (c-oncogenes) that encode proteins regulating growth, differentiation, and survival. They are NOT cancer genes — they are NORMAL genes performing essential physiological functions.
2. **Categories of products:** Growth factors (c-SIS → PDGF); Growth factor receptors with kinase activity (ERBB1/EGFR, HER2, RET, KIT); Intracellular signal transducers (RAS, RAF); Nuclear transcription factors (MYC, FOS, JUN); Cell cycle regulators (Cyclin D1, CDK4); Apoptosis regulators (BCL2).
3. **Regulatory roles in normal physiology:** Tightly controlled — expressed only when cell needs to proliferate (wound healing, development). Expression is transient, turned off by negative feedback, and regulated by growth factor signals.
4. **How they become oncogenes:** Any gain-of-function change — point mutation, amplification, translocation, insertional mutagenesis — converts the proto-oncogene (normal) to an oncogene (constitutively active). Only ONE allele needs to be altered (dominant).
5. **Key examples:** RAS proto-oncogene = GTPase switch. Oncogenic RAS = GTP-bound permanently (cannot hydrolyse GTP due to Gly-12 mutation). MYC = transcription factor for cell growth genes; translocated MYC in Burkitt's lymphoma = constitutively expressed under IgH promoter.
6. **Clinical relevance:** Targeting oncogenes has led to targeted cancer therapies — imatinib (BCR-ABL), trastuzumab (HER2), erlotinib (EGFR). Understanding proto-oncogene function is essential for drug design.

EQ – APOPTOSIS IS IMPORTANT FOR PREVENTING CANCER [PYQ 2023]

"Apoptosis is very important for preventing cancer."

1. **Definition of apoptosis:** Programmed cell death — genetically encoded, energy-dependent process. Characterised by cell shrinkage, chromatin condensation, DNA laddering (nucleosomal fragments of 180 bp), membrane blebbing, apoptotic bodies (phagocytosed — NO inflammation). Mediated by caspases.
2. **Normal anticancer role of apoptosis:** (a) p53 activation after DNA damage → if irreparable → BAX, PUMA, NOXA → mitochondrial pathway → cell dies before it can become cancerous. (b) Activated T cells kill virally infected or pre-malignant cells via Fas-FasL and perforin-granzyme. (c) c-MYC activation (e.g., from growth factor stimulation) normally triggers ARF pathway → p53-dependent apoptosis → prevents MYC-driven proliferation from going uncontrolled.
3. **How cancer cells evade apoptosis:** BCL2 overexpression (t(14;18) follicular lymphoma) → sequesters BAX, BIM → cannot trigger MOMP. p53 mutation (50% of cancers) → cannot activate apoptotic programme. Survivin/IAP family overexpression → inhibit caspases. Loss of Fas expression → evade CTL killing.
4. **Therapies targeting apoptosis:** Venetoclax (BCL2 inhibitor) → restores apoptosis in CLL/AML. BH3 mimetics → release BAX/BAK. Traditional chemotherapy agents work partly by triggering apoptosis in cancer cells. Immune checkpoint inhibitors → reactivate CTLs → Fas-FasL killing.
5. **Cytochrome c — dual role:** In healthy cells = electron carrier in ETC (Complex III → Complex IV). In apoptosis = released from mitochondria → apoptosome → caspase-9 activation. This dual role connects metabolism and cell death.
6. **Examiner tip:** Always mention both intrinsic (mitochondrial, caspase-9) and extrinsic (death receptor, caspase-8) pathways. In cancer prevention, intrinsic pathway (p53-BAX axis) is most important.

 EQ – INSULIN RECEPTOR HAS TYROSINE KINASE ACTIVITY [PYQ 2010-S]

"Insulin receptor has tyrosine kinase activity."

1. **Structure of insulin receptor:** Heterotetrameric glycoprotein — 2 α + 2 β subunits linked by S-S bonds ($\alpha\alpha$ and $\alpha\beta$). α subunits: extracellular, insulin binding site. β subunits: transmembrane + cytoplasmic, contain Tyr kinase domain. MW ~350 kDa (mature receptor).
2. **Evidence of Tyr kinase:** β -subunit cytoplasmic domain contains intrinsic tyrosine kinase activity (belongs to RTK superfamily). Kinase domain: DFG motif, activation loop Tyr (Y1158, Y1162, Y1163) — autophosphorylation of these Tyr residues activates kinase ~10-fold.
3. **Mechanism of activation:** Insulin binds α subunit → conformational change → transautophosphorylation of β subunit Tyr residues → kinase activated → phosphorylates IRS-1 (insulin receptor substrate-1) and Shc at Tyr residues → PI3K-AKT pathway (glucose uptake, glycogen synthesis, anti-apoptosis) + RAS-ERK pathway (gene expression, growth).
4. **GLUT4 translocation:** PI3K-AKT → AS160 (TBC1D4) phosphorylation → Rab-GTPase activation → GLUT4 vesicle fusion with plasma membrane → ↑ glucose uptake (muscle, adipose).
5. **Insulin resistance:** In Type 2 DM — receptor Tyr kinase activity impaired (serine phosphorylation of IRS-1 by inflammatory kinases like JNK, IKK → blocks Tyr signalling) → downstream pathway fails → hyperglycaemia despite insulin.
6. **Cancer connection:** IGF-1R (structurally similar to insulin receptor) → activates same PI3K-AKT pathway → promotes cancer cell survival and proliferation. Targeting IGF-1R = cancer therapy strategy.

17

Quickfire Revision Table



EXAM DAY HIGH-YIELD – MODULE 10

QUESTION	ANSWER
Only Ig that crosses placenta	IgG — via FcRn (neonatal Fc receptor) on syncytiotrophoblasts
First Ig in primary response	IgM — pentameric, best complement activator, best agglutinin
Secretory Ig in colostrum/saliva/tears	IgA (sIgA) — dimer with J chain + secretory component
Ig that triggers mast cell degranulation	IgE — binds FcεRI on mast cells. In allergy/anaphylaxis
Papain vs Pepsin cleavage products	Papain: 2 Fab + Fc (above hinge). Pepsin: F(ab') ₂ + pFc' (below hinge)
Type I hypersensitivity Ig	IgE. Mediators: histamine, leukotrienes (LTC ₄ /D ₄ /E ₄), PGD ₂ . Treatment: epinephrine
Type IV hypersensitivity — no Ig!	T cell mediated (CD4 ⁺ Th1 + macrophages). 48–72h delay. TB test, contact dermatitis
Most important complement fragment for opsonisation	C3b (binds CR1 on phagocytes). C3a + C5a = anaphylatoxins. MAC = C5b-9
BCR-ABL → drug target	Imatinib (Gleevec) — competitive inhibitor of BCR-ABL Tyr kinase. CML treatment
Philadelphia chromosome	t(9;22) → BCR-ABL fusion. CML. Best confirmed by FISH or RT-PCR
RAS mutation mechanism	Point mutation (G12V) → cannot hydrolyse GTP → RAS stuck "ON" → continuous proliferation. 30% of all cancers
p53 downstream effectors	Arrest: p21 (inhibits CDK2). Apoptosis: BAX, PUMA, NOXA. Repair: GADD45
Two-hit hypothesis — hereditary vs sporadic	Hereditary: 1st hit germline → bilateral, early. Sporadic: both hits somatic → unilateral, later. Classic example: Retinoblastoma (RB1)
HER2 in breast cancer — treatment	Trastuzumab (Herceptin) — blocks HER2 RTK. HER2 amplified in ~25% breast cancers
Aflatoxin B1 carcinogen → cancer	Hepatocellular carcinoma (HCC). From Aspergillus in mouldy peanuts. Activated by CYP1A2 → epoxide → DNA adducts → p53 codon 249 mutation
AFP marker for	HCC and Yolk sac germ cell tumours. Oncofetal antigen
CEA marker for	Colorectal cancer (monitoring recurrence). Also GI, lung, breast. Not specific enough for screening
CA-125 for	Epithelial ovarian cancer (monitoring). Also ↑ in endometriosis, PID
PSA organ-specific but not cancer-specific — why?	↑ in BPH, prostatitis too. Free/total PSA ratio helps discriminate prostate cancer
Warburg effect	Cancer cells use aerobic glycolysis (glucose → lactate even with O ₂). Basis of FDG-PET scan
Methotrexate mechanism	DHFR inhibitor → ↓ THF → ↓ TMP → ↓ DNA synthesis → S-phase block
Insulin receptor type	RTK (Receptor Tyrosine Kinase) — 2α + 2β subunits. β subunit has Tyr kinase domain
Cholera toxin mechanism	ADP-ribosylates Gas → permanently active → adenylyl cyclase always on → ↑↑↑ cAMP → CFTR → Cl ⁻ secretion → diarrhoea

cGMP second messenger — example

ANP/BNP → membrane guanylyl cyclase → cGMP → PKG → vasodilation. Also NO → soluble GC → cGMP

MODULE 10 MASTER MNEMONICS

IG CLASSES ORDER

"GAMED" = IgG (most), IgA (mucosal), IgM (macro), IgE (eosinophil/allergy), IgD (differentiation)

HYPERSENSITIVITY TYPES

"ACID" = Anaphylactic (I), Cytotoxic (II), Immune complex (III), Delayed (IV)

TUMOUR MARKERS

"AFP-HCC, CEA-Colon, PSA-Prostate, CA125-Ovary, CA19-9-Pancreas, hCG-Choriocarcinoma"

COMPLEMENT FUNCTIONS

"CLAMO" = Chemotaxis (C5a), Lysis (MAC), Anaphylatoxin, Mast cell, Opsonisation (C3b)

MODULE SERIES COMPLETE

Modules 01–10 of the Caffeine & Cadaver MBBS Biochemistry Note Series are now complete. This module (10) covers Immunochemistry, Oncogenesis, Signal Transduction, and Hormone Receptors — PYQ-referenced against MBBS 2010–2025 throughout.

Modules covered: 01 Enzyme Inhibition · 02 Carbohydrate Metabolism · 03 Lipid Metabolism · 04 Protein Chemistry & Haemoglobin · 05 Nucleotide Chemistry · 06 Molecular Biology & Genetics · 07 Biological Oxidation · 08 Nutrition, Vitamins & Minerals · 09 Clinical Biochemistry · 10 Immunochemistry & Oncogenesis

MODULE 10 · IMMUNOCHEMISTRY & ONCOGENESIS

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